1 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`
.

1.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 4
- Solver accepted/rejected steps: 416 / 1
- RHS evaluations: 104426
- Solver tolerances: abstol = 1.000×10^{-8}, reltol = 1.000×10^{-6}

1.2 Forcing and Boundary Conditions

Table 1: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	$8.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	$265.000000\,\mathrm{K}$
Deep soil temperature	T_{deep}	$262.000000\,\mathrm{K}$
Downward longwave forcing	$R_{\downarrow}$	$245.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	$0.100000\,\mathrm{m}$
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	$265.000000\,\mathrm{K}$
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

1.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.103721, 79.407743]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.009840, 82.319866]$

1.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

1.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

1.6 Included Plots

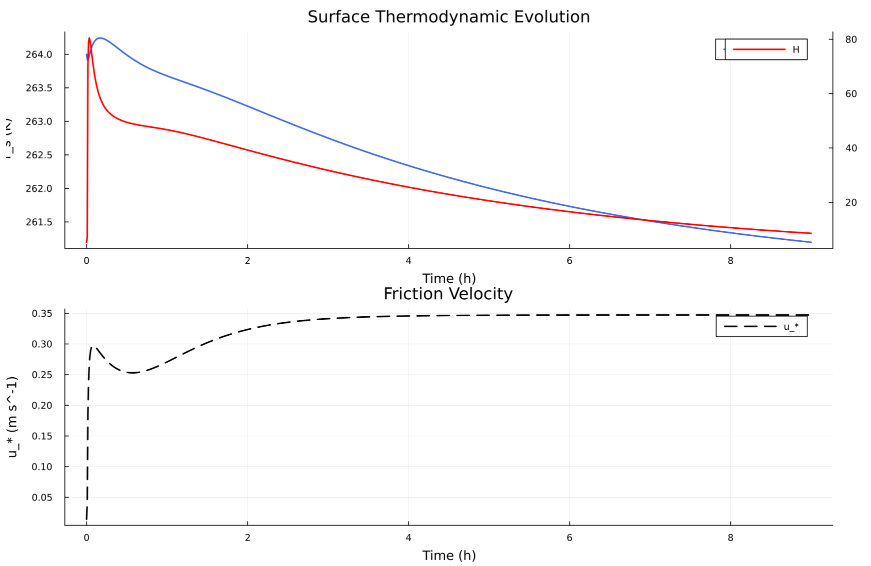

Figure 1: Time series of T__s, sensible heat flux, and friction velocity

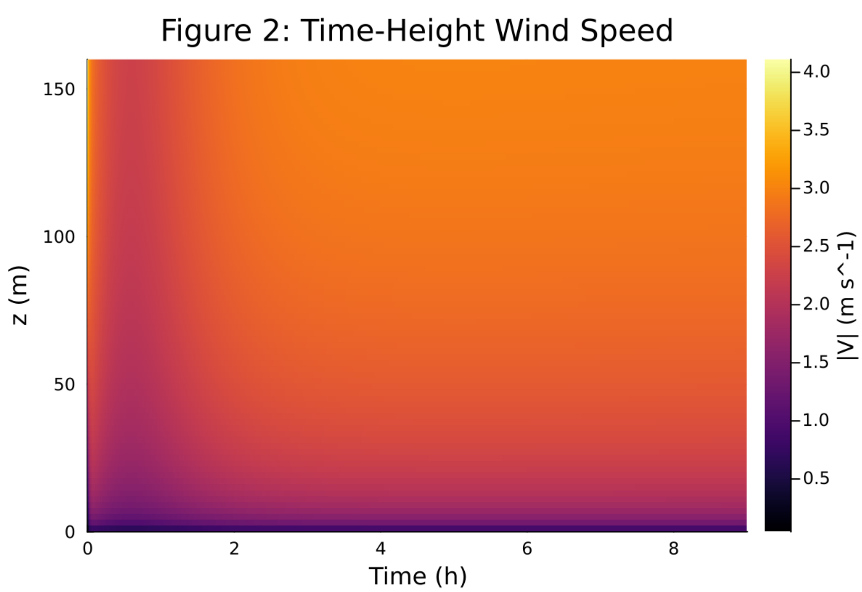

Figure 2: Time-height contour of wind speed with LLJ evolution

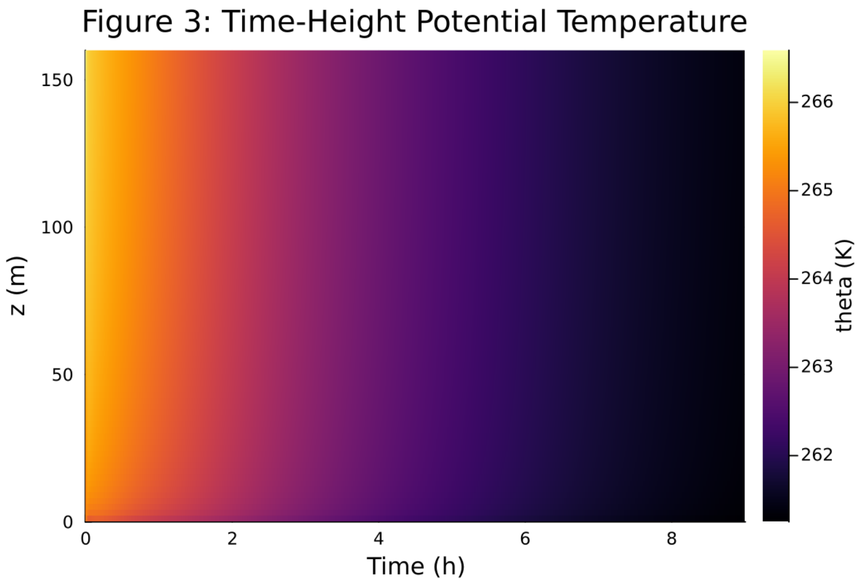

Figure 3: Time-height contour of potential temperature and inversion growth

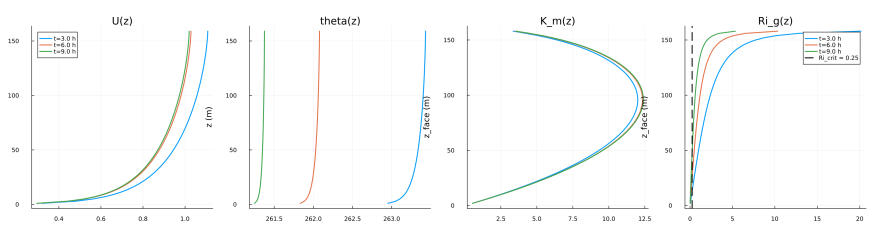

Figure 4: Vertical profiles of U, theta, and Km at representative times

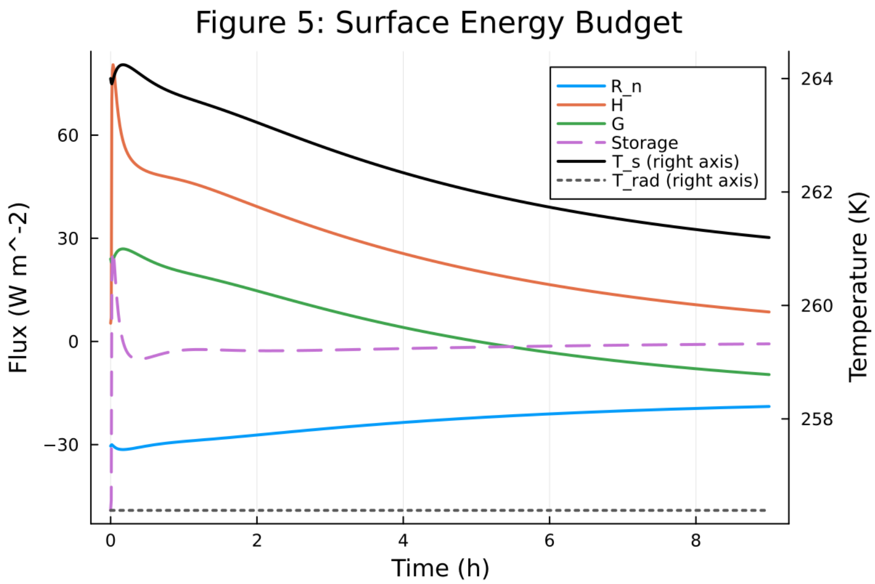

Figure 5: Surface energy budget components (R__n, H, G, storage)

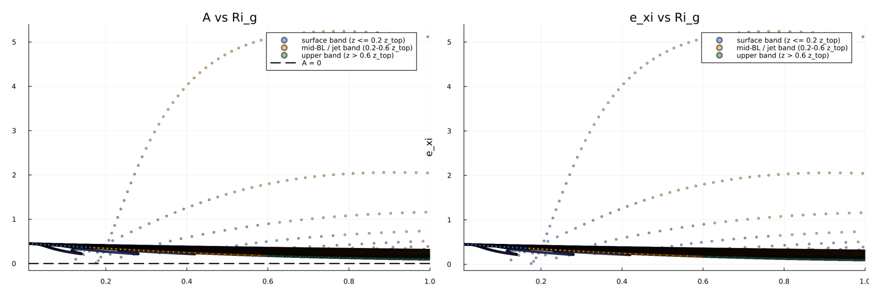

Figure 6: Phase portrait on regularized manifold (Delta vs e__xi)

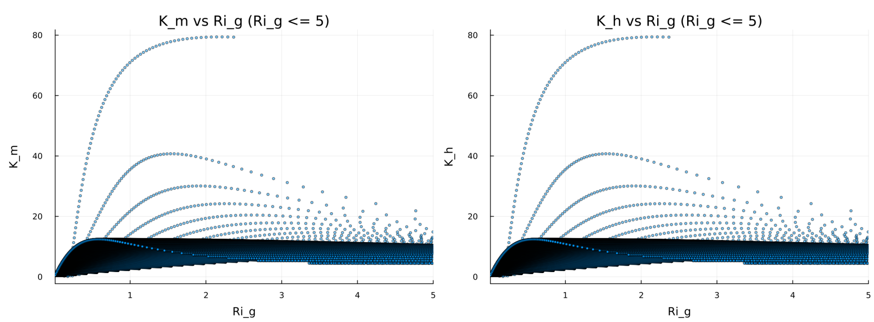

Figure 7: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

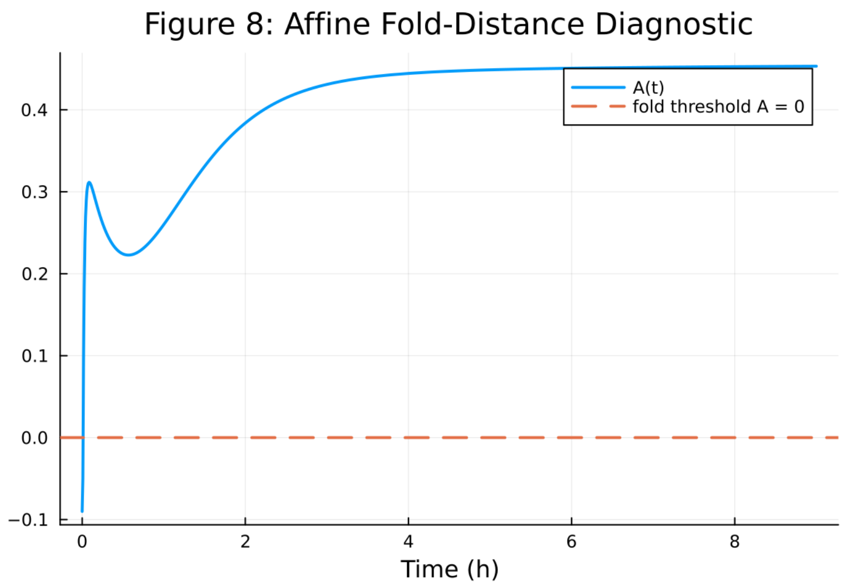

Figure 8: Fold proximity diagnostics versus time